



VMamba: Visual State Space Model

Liu, Yue, et al. "Vmamba: Visual state space model." Advances in neural information processing systems 37 (2024)

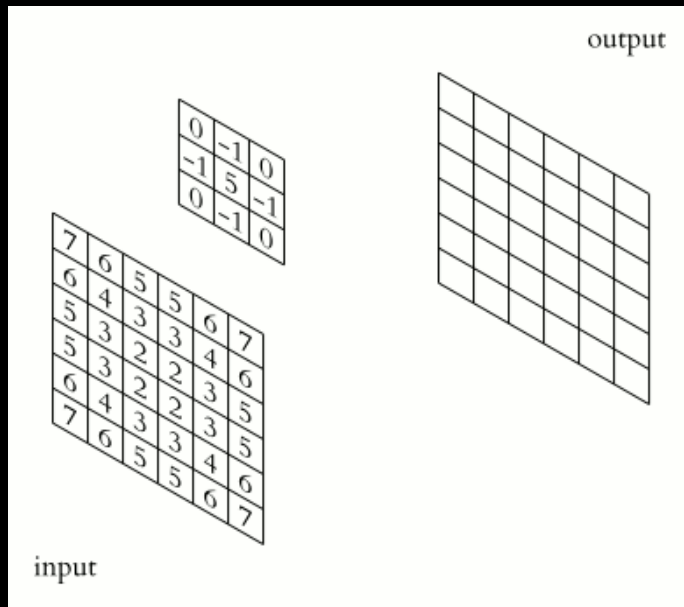
허정원 이미지처리팀

Contents

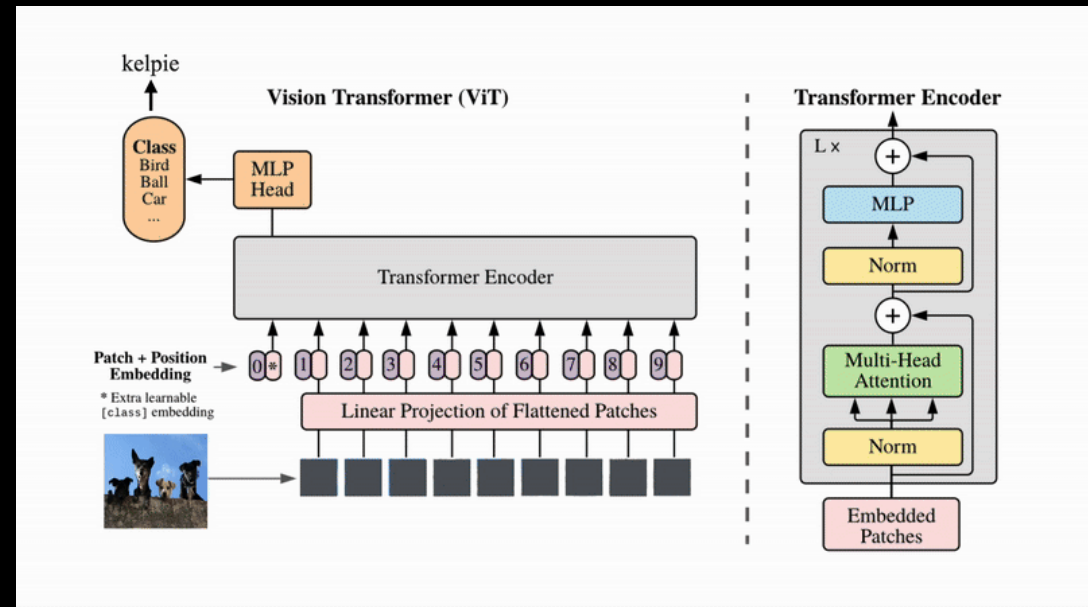
- **Introduction**
- **Related Work**
- **VMamba**
- **Experiments**
- **Conclusion**

Introduction – CNN/ViT

- To represent complex patterns in vision data, two primary categories of backbone networks, i.e., **CNNs** and **ViTs**, have been proposed and utilized in visual tasks.



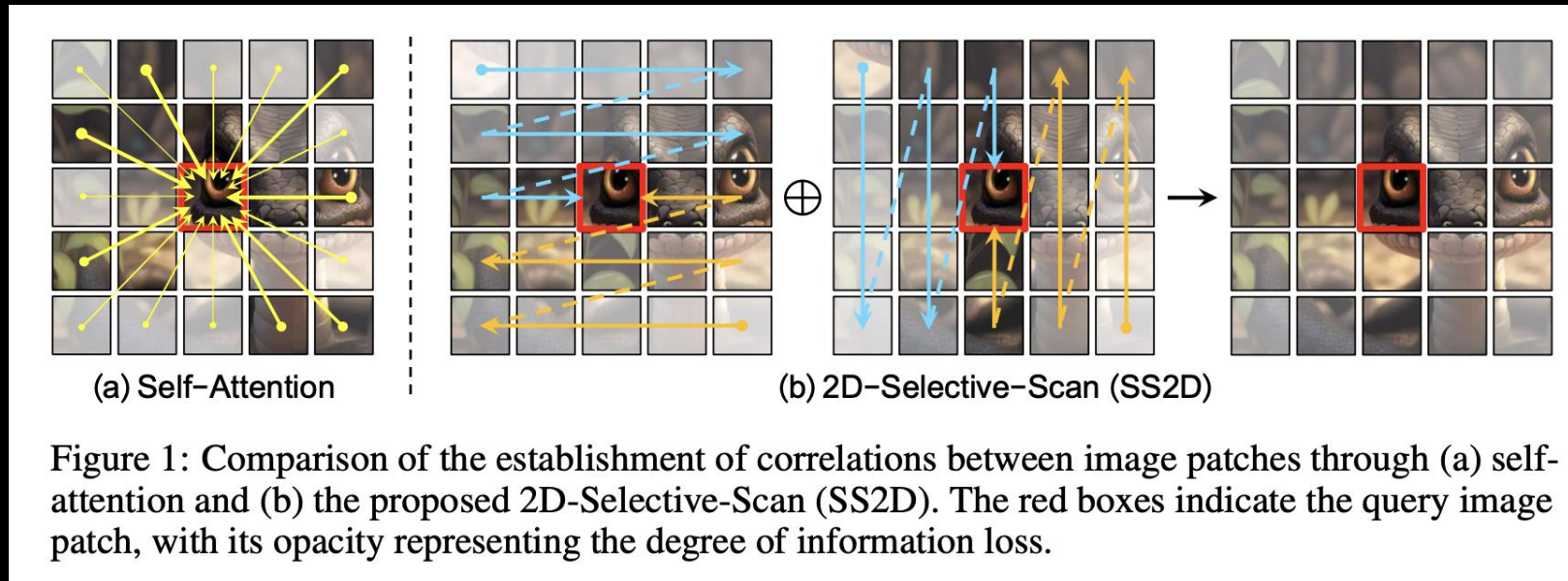
Convolutional Neural Networks



Vision Transformers

- However, the **quadratic** complexity of self-attention w.r.t. the number of tokens imposes substantial computational overhead in downstream tasks involving **large spatial resolutions**.

Introduction – VMamba

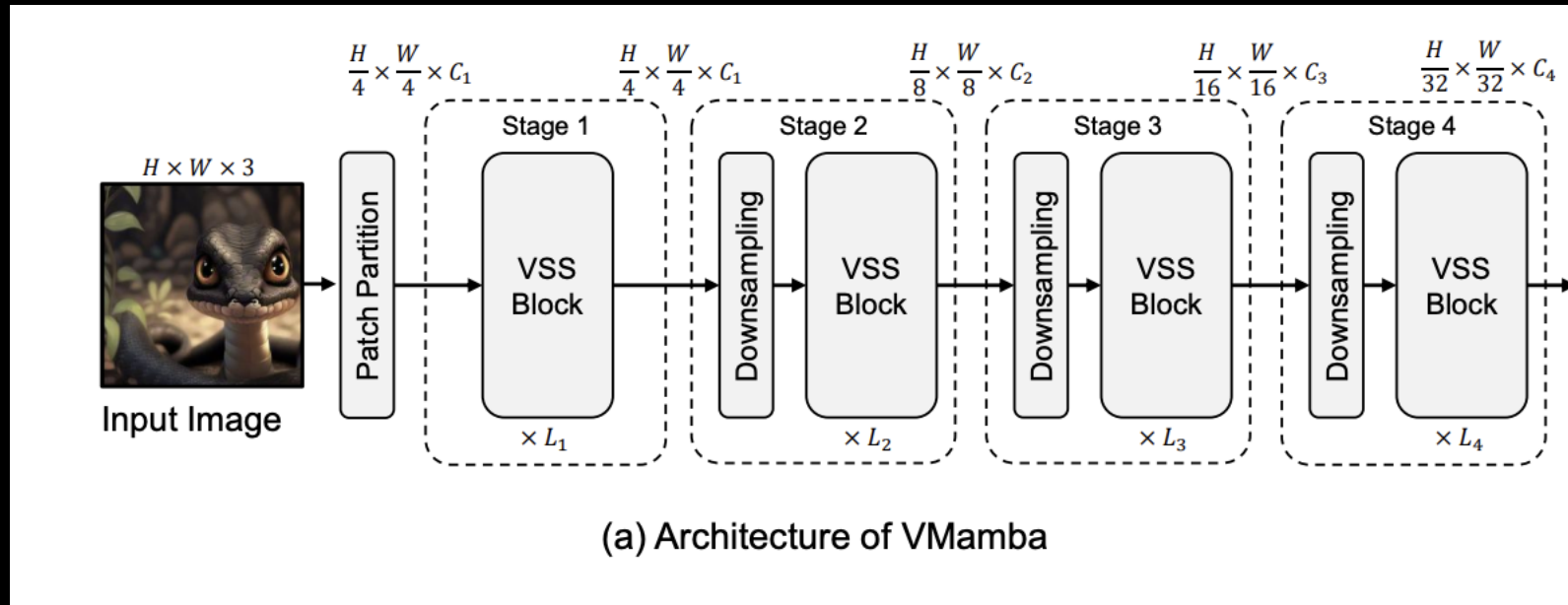


- Recently, **Mamba**, an innovative **State Space Model (SSM)** in the field of natural language processing (NLP), has emerged as a promising approach for long-sequence modeling with linear complexity.
- To address the challenge of processing vision data, we propose **2D Selective Scan (SS2D)**, a four-way scanning mechanism designed for spatial domain traversal, thereby reducing the computational complexity from quadratic to linear.

Related Work

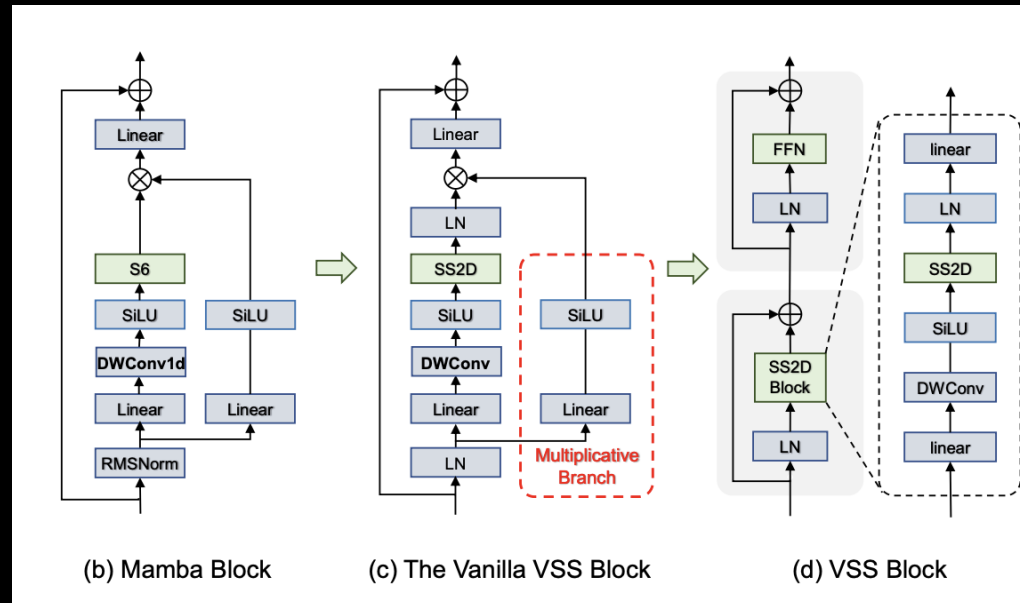
- Convolutional Neural Networks Since **AlexNet**, considerable efforts have been devoted to enhancing the modeling capabilities and computational efficiency of CNN-based models across various visual tasks.
- Vision Transformers (ViTs) As a pioneering work, ViT explores the effectiveness of vision models based on **vanilla Transformer** architecture, highlighting the importance of large-scale pre-training for **image classification** performance.
- State Space Models (SSMs) In efforts to improve scaling efficiency, SSMs have emerged as compelling **alternatives to Transformers**, attracting significant research attention especially when handling long input sequences.

Vmamba



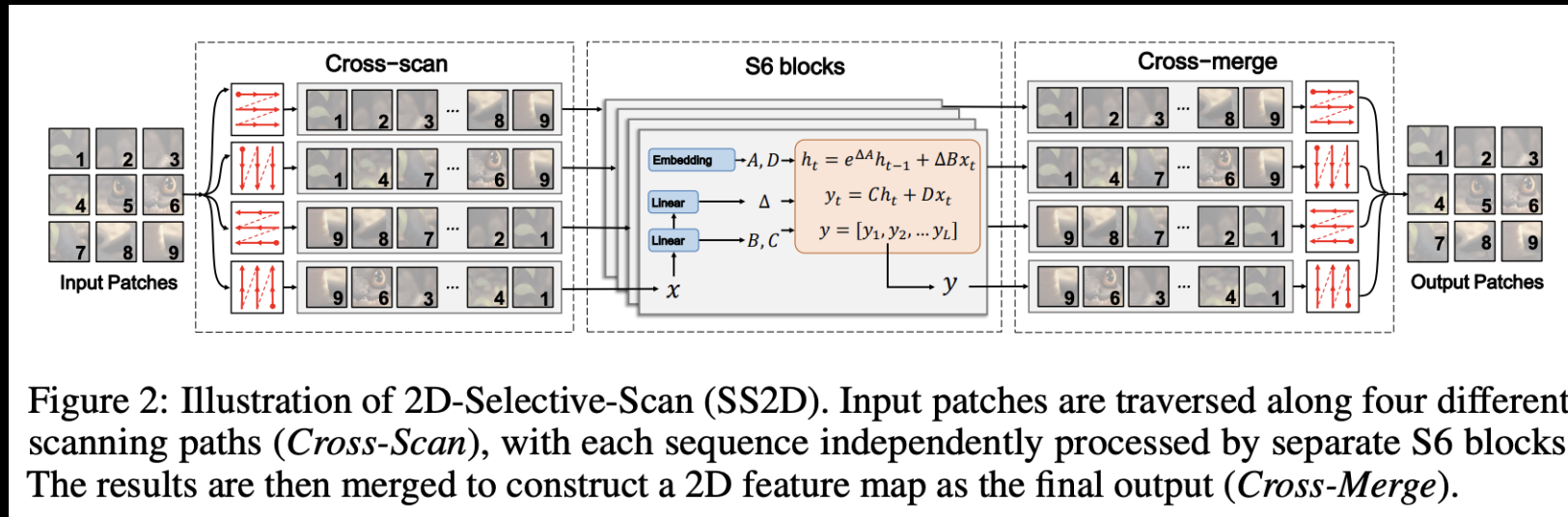
- We develop VMamba in three scales: Tiny, Small, and Base, where multiple network stages are employed to create hierarchical representations, and specifically, each stage comprises a down-sampling layer followed by a stack of **Visual State Space (VSS)** blocks

Vmamba - VSS



- **The VSS blocks** serve as the visual counterparts to Mamba blocks, and the improved VSS block consists of **a single network branch** with two residual modules, mimicking the architecture of a **vanilla Transformer** block.
- We substitute the S6 (Structured State Space Sequence + Selective Scan) module with the newly proposed 2D-Selective-Scan (SS2D) module to facilitate the collection of contextual information from non-sequential 2D vision data.

Vmamba – SS2D



Step 1: Cross-Scan Unfolds the input patches into sequences along **four distinct traversal paths**.

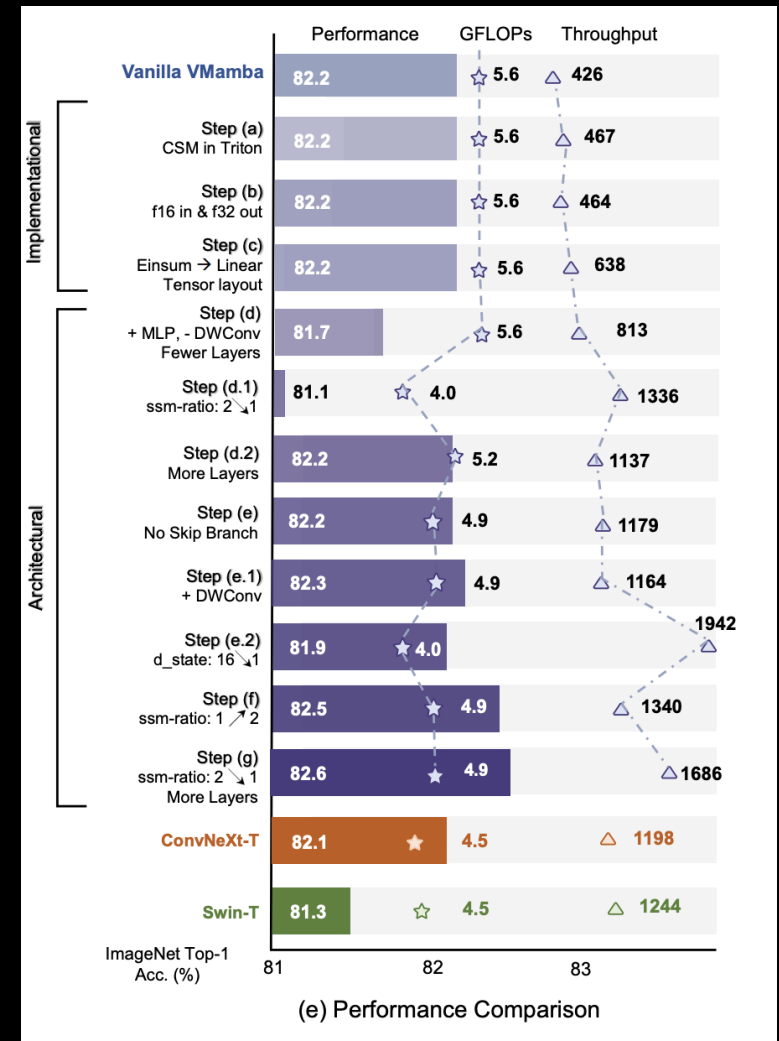
Step 2: Selective Scanning with S6 Blocks Processes each patch sequence in parallel using a separate S6 block.

Step 3: Cross-Merge Reshapes and merges the resultant sequences to form the output map.

=> Integrates information from all pixels across different directions to establish global receptive fields in 2D space.

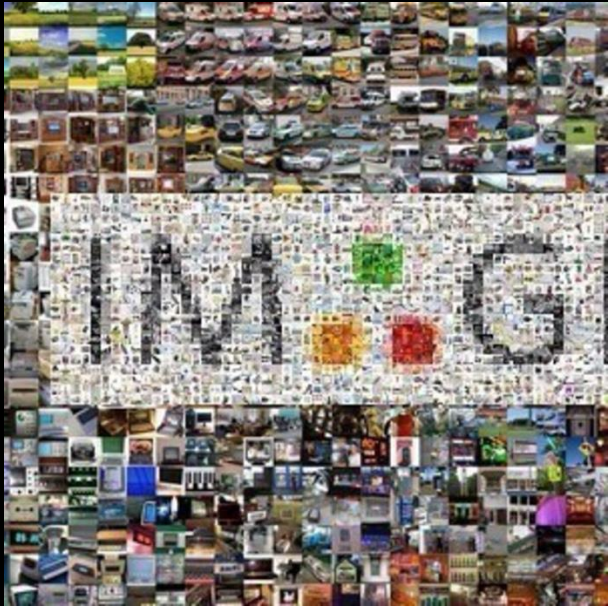
Vmamba — Accelerating VMamba

- Step (e) by reducing the parameter ssm-ratio (the feature expansion factor) from 2.0 to 1.0 (also referred to as Step (d.1)), raising the layer numbers to [2,2,5,2] (also referred to as Step (d.2)), and discarding the entire multiplicative branch.
- Step (f) by introducing the **DWConv** layers (also referred to as Step (e.1)) and reducing the parameter d_state (the SSM state dimension) from 16.0 to 1.0 (also referred to as Step (e.2)), together with raising ssm-ratio back to 2.0.

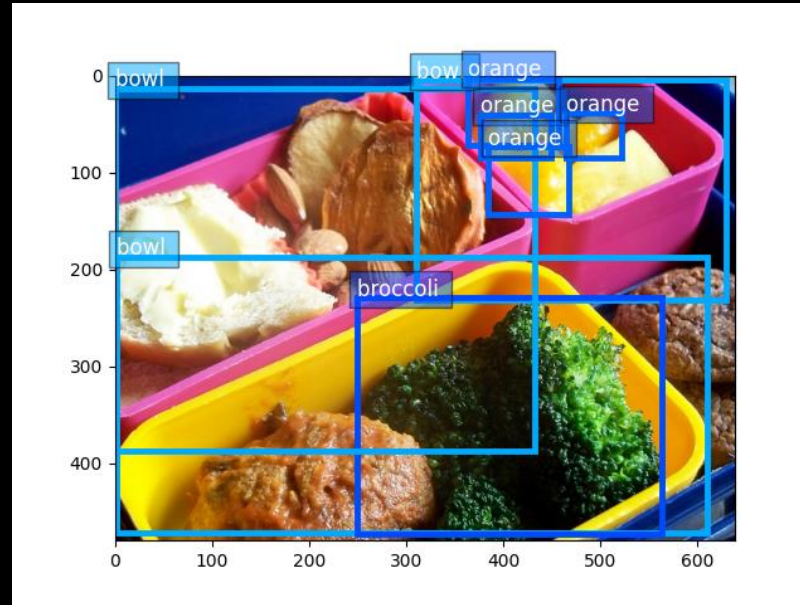


Experiments

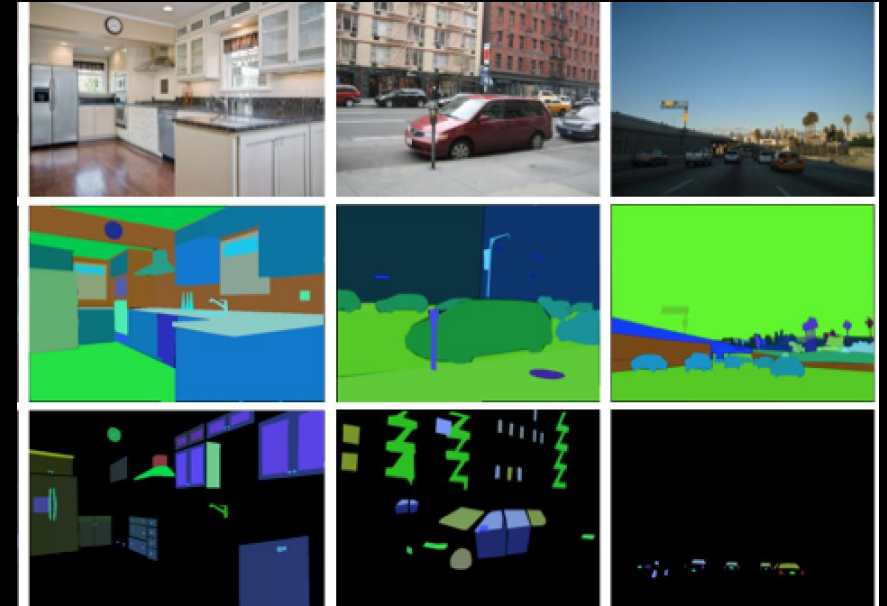
- We evaluate VMamba's performance in **image classification on ImageNet-1K**, **object detection and instance segmentation on MSCOCO2017**, and **semantic segmentation on ADE20K**.
- Compared to benchmark vision models built on **CNNs** (ConvNeXt), **ViTs** (Swin, HiViT), and **SSMs** (S4ND, Vim).



ImageNet 1K



MSCOCO 2017



ADE 20K

Experiments – ImageNet 1k

Table 1: Performance comparison on ImageNet-1K. Throughput values are measured with an A100 GPU and an AMD EPYC 7542 CPU, using the toolkit released by [62], following the protocol proposed in [36]. All images are of size 224×224 .

Model	Params (M)	FLOPs (G)	TP. (img/s)	Top-1 (%)	Model	Params (M)	FLOPs (G)	TP. (img/s)	Top-1 (%)
Transformer-Based					ConvNet-Based				
DeiT-S [57]	22M	4.6G	1761	79.8	ConvNeXt-T [37]	29M	4.5G	1198	82.1
DeiT-B [57]	86M	17.5G	503	81.8	ConvNeXt-S [37]	50M	8.7G	684	83.1
HiViT-T [66]	19M	4.6G	1393	82.1	ConvNeXt-B [37]	89M	15.4G	436	83.8
HiViT-S [66]	38M	9.1G	712	83.5	SSM-Based				
HiViT-B [66]	66M	15.9G	456	83.8	S4ND-Conv-T [44]	30M	5.2G	683	82.2
Swin-T [36]	28M	4.5G	1244	81.3	S4ND-ViT-B [44]	89M	17.1G	397	80.4
Swin-S [36]	50M	8.7G	718	83.0	Vim-S [69]	26M	5.3G	811	80.5
Swin-B [36]	88M	15.4G	458	83.5	VMamba-T	30M	4.9G	1686	82.6
XCiT-S24 [1]	48M	9.2G	671	82.6	VMamba-S	50M	8.7G	877	83.6
XCiT-M24 [1]	84M	16.2G	423	82.7	VMamba-B	89M	15.4G	646	83.9

- With similar FLOPs, VMamba-T achieves a top-1 accuracy of 82.6%, outperforming DeiT-S by 2.8% and Swin-T by 1.3%.
- Notably, VMamba maintains its performance advantage at both Small and Base scales. For example, VMamba-B achieves a top-1 accuracy of 83.9%, surpassing DeiT-B by 2.1% and Swin-B by 0.4%.
- In terms of computational efficiency, VMamba-T achieves a throughput of 1,686 images/s, which is either superior or comparable to state-of-the-art methods.

Experiments - COCO

- VMamba demonstrates superior performance in both box and mask Average Precision (AP^b and AP^m) across different training schedules.
- Under the 12-epoch fine-tuning schedule, VMamba-T/S/B achieves object detection mAPs of 47.3%/48.7%/49.2%, outperforming Swin-T/S/B by 4.6%/3.9%/2.3% mAP.
- For semantic segmentation on ADE20K, VMamba-T achieves 3.4% higher mIoU than Swin-T and 1.9% higher than ConvNeXt-T in the Single-Scale (SS) setting.

Table 2: **Left:** Results for object detection and instance segmentation on MSCOCO. AP^b and AP^m denote box AP and mask AP, respectively. FLOPs are calculated with an input size of 1280×800 . The notation ‘ $1 \times$ ’ indicates models fine-tuned for 12 epochs, while ‘ $3 \times MS$ ’ denotes multi-scale training for 36 epochs. **Right:** Results for semantic segmentation on ADE20K. FLOPs are calculated with an input size of 512×2048 . ‘SS’ and ‘MS’ denote single-scale and multi-scale testing, respectively.

Mask R-CNN $1 \times$ schedule					ADE20K with crop size 512				
Backbone	AP^b	AP^m	Params	FLOPs	Backbone	mIOU (SS)	mIOU (MS)	Params	FLOPs
Swin-T	42.7	39.3	48M	267G	ResNet-50	42.1	42.8	67M	953G
ConvNeXt-T	44.2	40.1	48M	262G	DeiT-S + MLN	43.8	45.1	58M	1217G
VMamba-T	47.3	42.7	50M	271G	Swin-T	44.5	45.8	60M	945G
Swin-S	44.8	40.9	69M	354G	ConvNeXt-T	46.0	46.7	60M	939G
ConvNeXt-S	45.4	41.8	70M	348G	NAT-T	47.1	48.4	58M	934G
VMamba-S	48.7	43.7	70M	349G	Vim-S	44.9	-	46M	-
Swin-B	46.9	42.3	107M	496G	VMamba-T	47.9	48.8	62M	949G
ConvNeXt-B	47.0	42.7	108M	486G	ResNet-101	43.8	44.9	86M	1030G
VMamba-B	49.2	44.1	108M	485G	DeiT-B + MLN	45.5	47.2	144M	2007G
Mask R-CNN $3 \times MS$ schedule					Swin-S	47.6	49.5	81M	1039G
Swin-T	46.0	41.6	48M	267G	ConvNeXt-S	48.7	49.6	82M	1027G
ConvNeXt-T	46.2	41.7	48M	262G	NAT-S	48.0	49.5	82M	1010G
NAT-T	47.7	42.6	48M	258G	VMamba-S	50.6	51.2	82M	1028G
VMamba-T	48.8	43.7	50M	271G	Swin-B	48.1	49.7	121M	1188G
Swin-S	48.2	43.2	69M	354G	ConvNeXt-B	49.1	49.9	122M	1170G
ConvNeXt-S	47.9	42.9	70M	348G	NAT-B	48.5	49.7	123M	1137G
NAT-S	48.4	43.2	70M	330G	RepLNet-31B	49.9	50.6	112M	1170G
VMamba-S	49.9	44.2	70M	349G	VMamba-B	51.0	51.6	122M	1170G

Experiments - Adaptability & Scalability

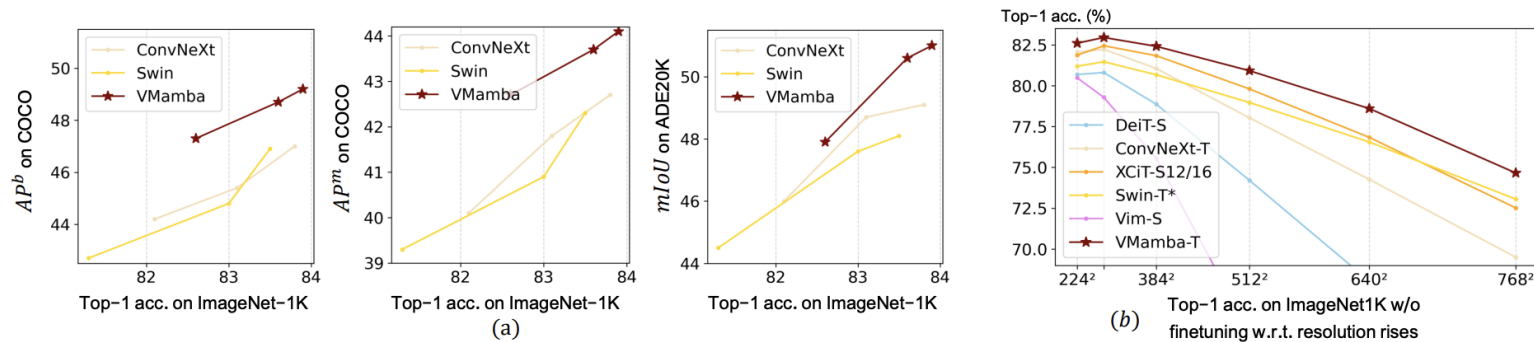


Figure 4: Illustration of VMamba’s adaptability to (a) downstream tasks and (b) input images with progressively increasing resolutions. Swin-T* denotes Swin-T tested with scaled window sizes.

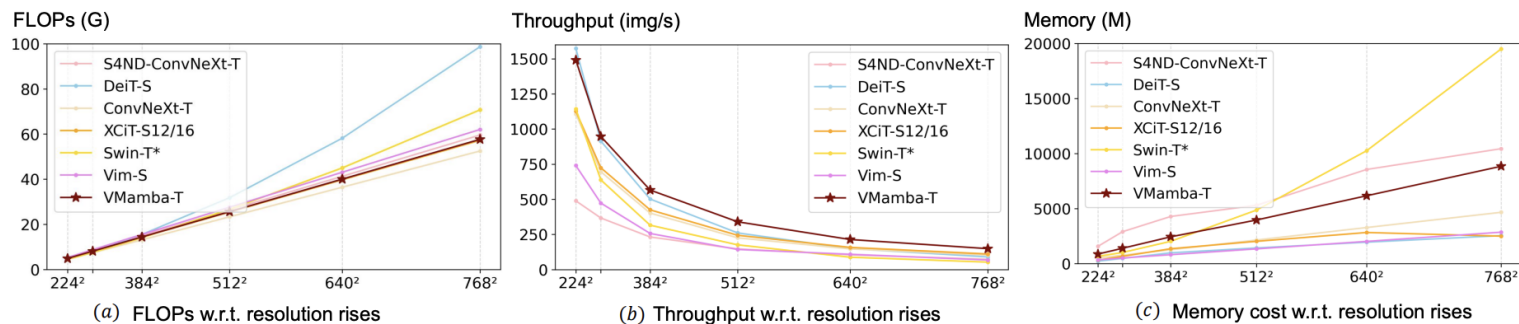
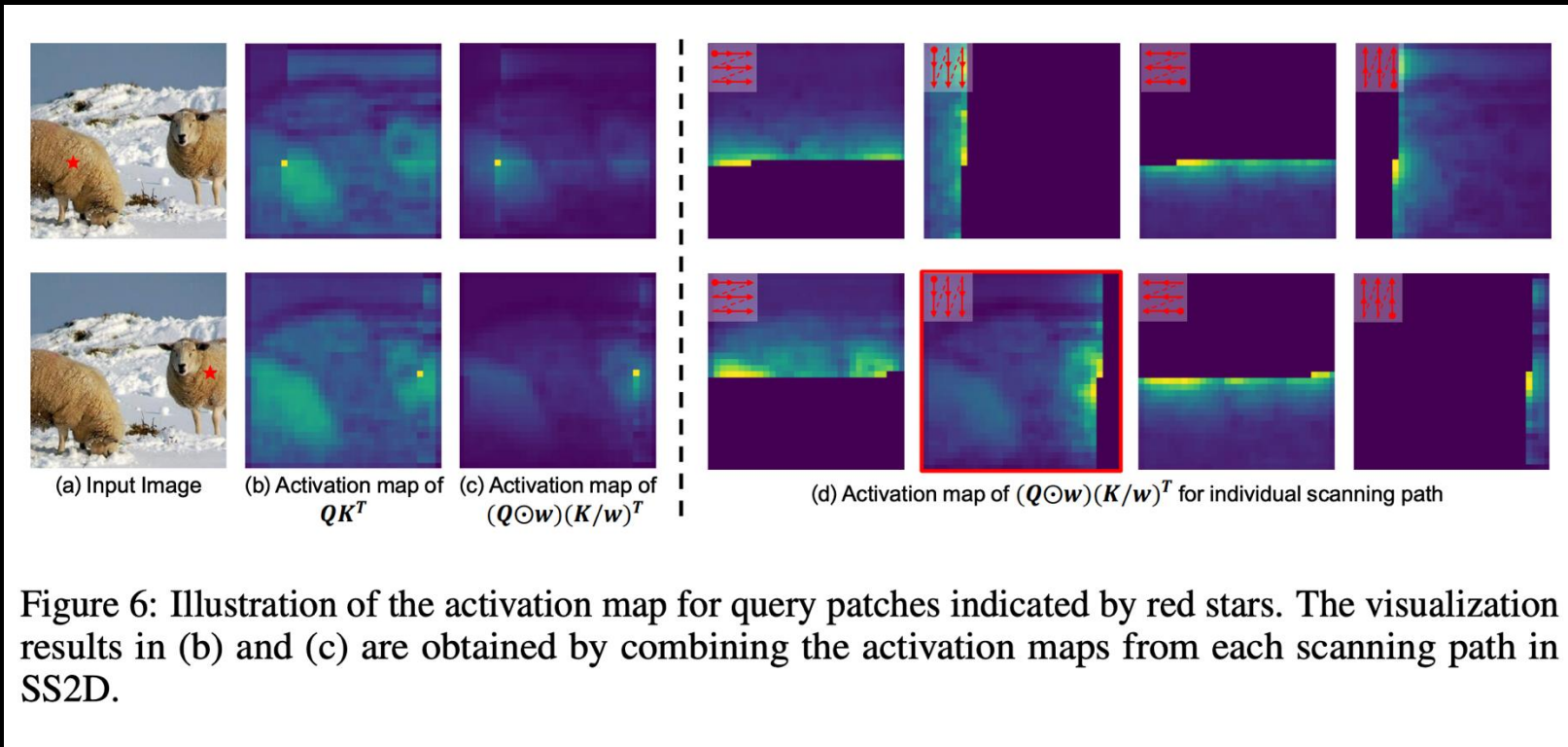


Figure 5: Illustration of VMamba’s resource consumption with progressively increasing resolutions. Swin-T* denotes Swin-T tested with scaled window sizes.

Experiments -Analysis



- Global Context Capture (QK^T)
- Selective Focus with Weighting (w)

Conclusion

- This paper presents **VMamba**, an efficient vision backbone model built with State Space Models (SSMs).
- VMamba integrates the advantages of selective SSMs from NLP tasks into visual data processing, bridging the gap between ordered 1D scanning and non-sequential 2D traversal through the novel **SS2D module**.
- Furthermore, we have **significantly improved the inference speed** of VMamba through a series of architectural and implementation refinements.
- The effectiveness of the VMamba family has been demonstrated through extensive experiments, and **its linear time complexity** makes VMamba advantageous for downstream tasks with **large-resolution inputs**.

Questions?